



AG00686 BMC 6Cyln.

ACC-Drive CRK-ø160; AC-ø127

Tensioner T38624; CW; 24.6Nm

8PK1475HD Fleetranner

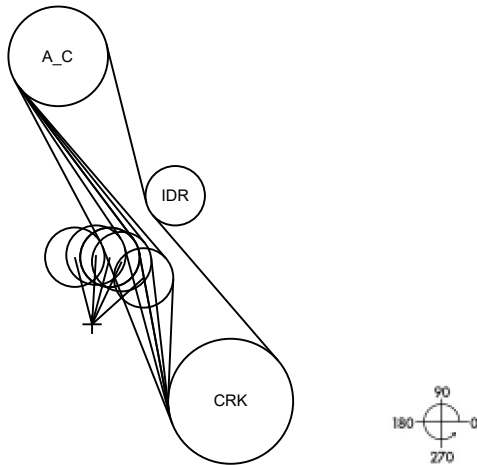
07Sep2023

Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016

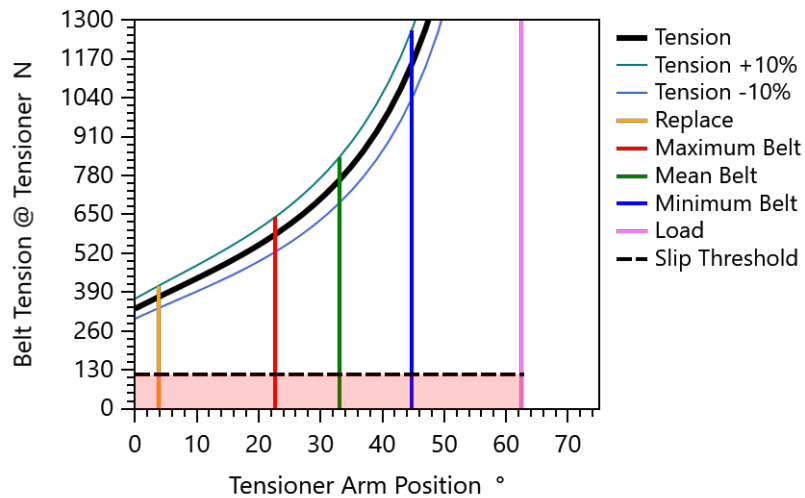
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Effective Belt Length mm (ISO 9981)	1475

Tensioner Data

Type	Automatic
Design Tension N	766 (172 lbf)



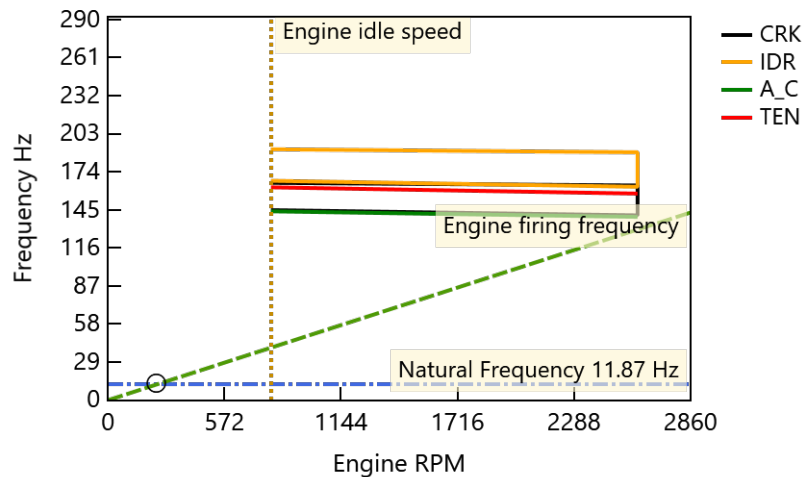
Belt Tension Control



Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2p-MT3 hours
95	1	10540

Natural Frequency Map



Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °
CRK	1262	1000	1113	1816.5	119 / 210
IDR	1262	1000	1110	512.6	28 / 27
A_C	1262	1000	810	1883.0	294 / 203
TEN	800	1000	766	362.0	211 / 26

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)

Permissible Grooved Pulley

Axial Offset Allowance

For 0.90° Belt Entry Angle

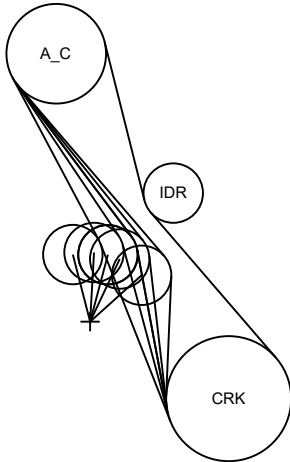
Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
A_C->CRK	IDR (0.33)	3.93	74	26
CRK->A_C	TEN (1.20)	1.49	7	93



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Layout Data mm

	X	Y	Flat	Pitch	Effective	DOB
CRK	0.00	0.00		162.40	160.00	160.99
IDR	-72.00	267.00	75.00	77.20	79.60	
A_C	-224.00	448.00		129.40	127.00	127.99
TEN	-156.85	186.97	75.00	77.20	79.60	



Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016

Belt Code	MT610
# of Ribs / Cord Material	8 / polyester
Stretch and Wear Allowance % of Length	0.70
Belt Length Tolerance (+/-) mm	6.00

Tensioner Data

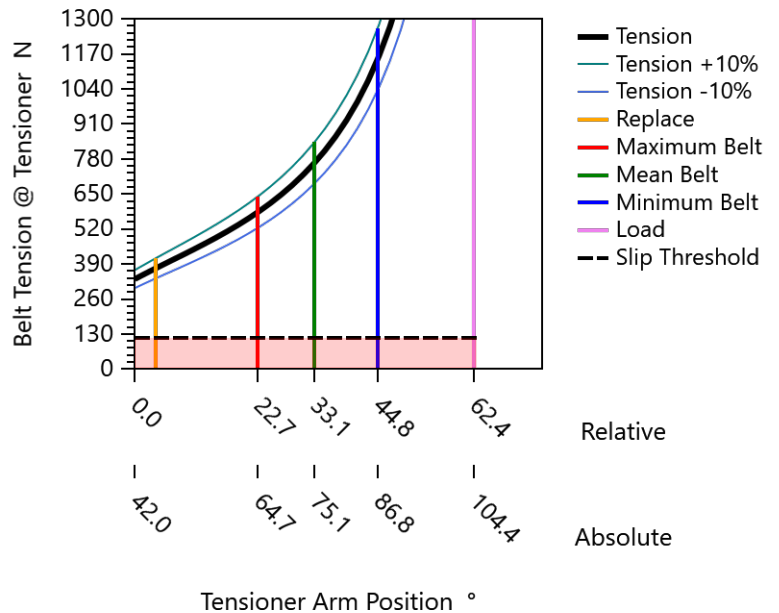
Type	Automatic
Design Tension N	766 (172 lbf)
Pivot Point {X, Y Coordinates} mm	-180.00 , 100.00
Arm Length mm	90.0
Spring Mean Load Nm	24.54
Spring Pre-Load Nm	8.59
Spring Rate Nm/deg	0.482

RESULTS

Belt Drive System Geometry

	Span Length mm	Wrap Angle °	Speed Ratio (Ref. Engine)
CRK	249.2	210.2	1.000
IDR	212.6	26.7	2.104
A_C	248.9	202.9	1.255
TEN	212.6	26.4	2.104

Belt Tension Control





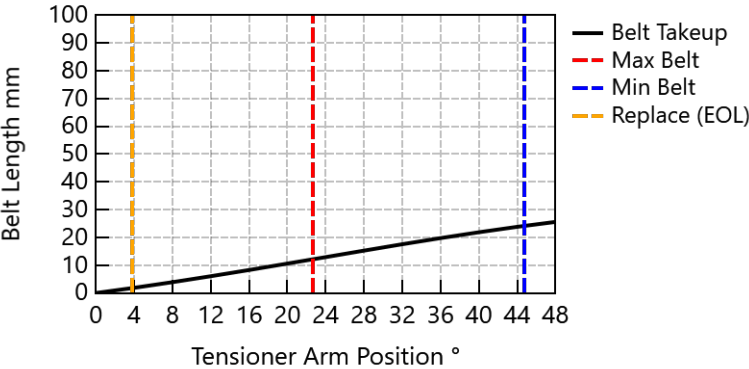
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Tensioner Geometry

Belt Takeup / Tensioner Arm Ratio mm/° 0.559

Position Description	Belt-Drive Length					
	Free Arm	Replace	Max	Mean	Min	Load
Arm Position °	42.0	45.8	64.7	75.1	86.8	104.4
Idler X Coordinate mm	-113.1	-117.2	-141.5	-156.8	-174.9	-202.4
Idler Y Coordinate mm	160.2	164.5	181.4	187.0	189.9	187.2
Hubload to Arm Angle °	22.9	26.0	41.9	51.3	62.3	79.7
Hubload Direction °	199.1	199.8	202.7	203.8	204.5	204.7
Hubload N	245.3	264.5	324.6	349.3	378.5	436.8
Tension N	334.2	373.5	583.0	765.7	1151.6	3863.1
Idler Wrap Angle °	43.1	41.5	32.3	26.4	18.9	6.5
Effective Drive Length mm	1496.7	1494.9	1484.5	1478.5	1472.5	1466.7
Required Eff. Belt Length mm	1493.2	1491.4	1481.0	1475.0	1469.0	1463.3

Belt Take-up





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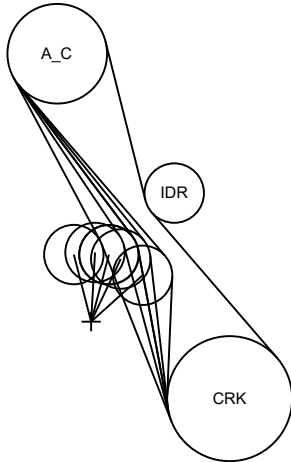
Tensioner T38624; CW; 24.6Nm

8PK1475HD Fleetranner

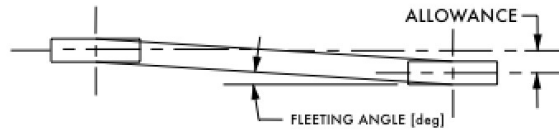
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Alignment Data °

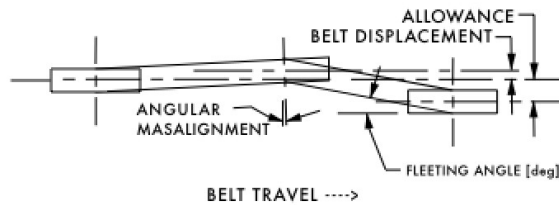
Fleeting Angle	0.90
IDR Angular Misalignment	0.33
TEN Angular Misalignment	1.20



(GROOVED - GROOVED) PULLEYS



(GROOVED - FLAT - GROOVED) PULLEYS



RESULTS

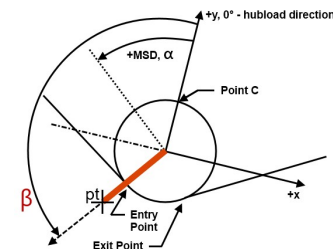
Permissible Grooved Pulley

Axial Offset Allowance

For 0.90° Belt Entry Angle

Adjacent Grooved Pulleys	Flat Pulley Angular Misalignment °	Axial Offset Allowance mm	Grooved Pulley Axial Offset %	Attributed to Flat Pulley Angular Misalignment %
A_C->CRK	IDR (0.33)	3.93	74	26
CRK->A_C	TEN (1.20)	1.49	7	93

Hubload to Arm Angle & MSD [°]



Hubload-to-arm angle is the angle between hubload and arm where arm direction is from pulley center to pivot.

β = Hubload-to-arm angle
pt = Tensioner pivot point

Backside Pulley Alignment Sensitivity

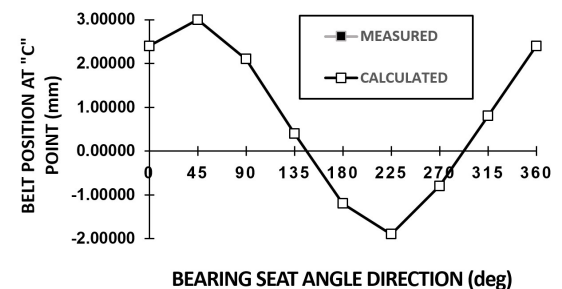
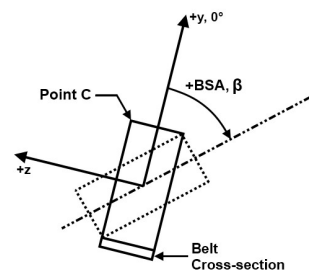
Backside Pulley	Angular Misalignment	Most Sensitive Direction	Resulting Fleeting Angle	Suggestions
2: IDR	0.333	69.96	0.23	AB C E
4: TEN	1.200	70.02	0.84	AB C E

Suggestion List for Backside Pulley Alignment Sensitivity

- A] Increase EXIT span length
- B] Decrease ENTRANCE span length
- C] Increase both ENTRANCE / EXIT span lengths
- D] Increase wrap angle
- E] Decrease wrap angle

Note: Most Sensitive Direction for Grooved pulleys is parallel to the belt entry span.

Direction of Bearing Seat Angle [°]

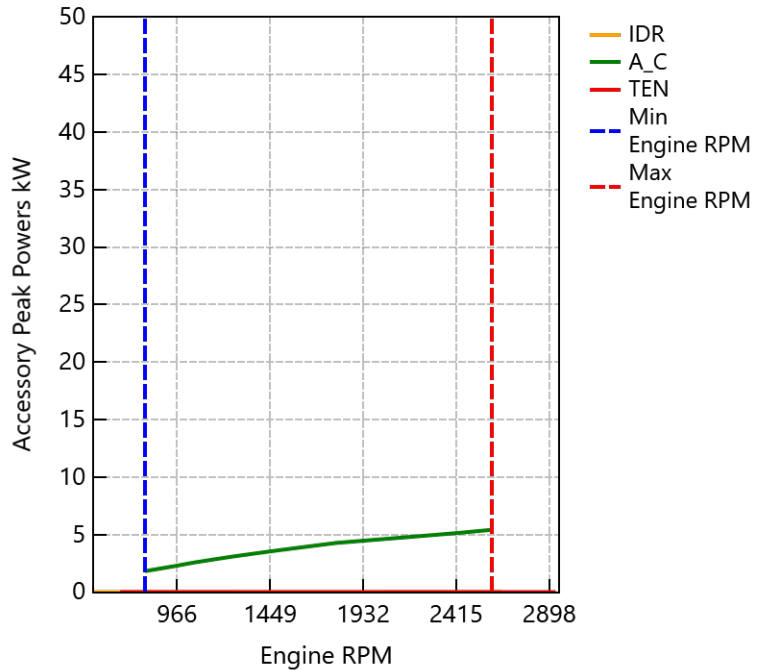




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Design Tension N	766 (172 [lbf])
Engine Acceleration RPM/s	1000
Engine Deceleration RPM/s	1000

Accessory Peak Powers kW



Accessory Data

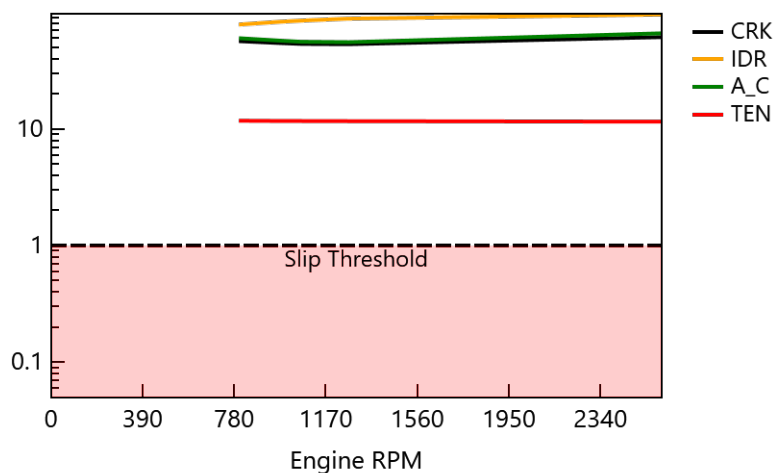
	File Name - if used	Moment of Inertia kg m ²
IDR	IDR.cmp	0.0004
A_C	A_C.cmp	0.0050
TEN	TEN.cmp	0.0076

RESULTS

Belt Slip Sensitivity

Pulley	Most Critical Load Condition					Required Increase Tension or Wrap Angle	
	Acceler. RPM/s	Load Combination %	A_C			N	°
CRK	1000	100					
IDR	1000	10					
A_C	-1000	100					
TEN	-1000	100					

Belt Slip Safety Factor





User: Abdurrahman Kol

Design: AG00686_BMC_6Cyln_ACC-Drive_CRKø160-ACø127_8PK1475HD_T38624-24.6Nm_CW_07Sep2023

AG00686 BMC 6Cyln.

ACC-Drive CRK-ø160; AC-ø127

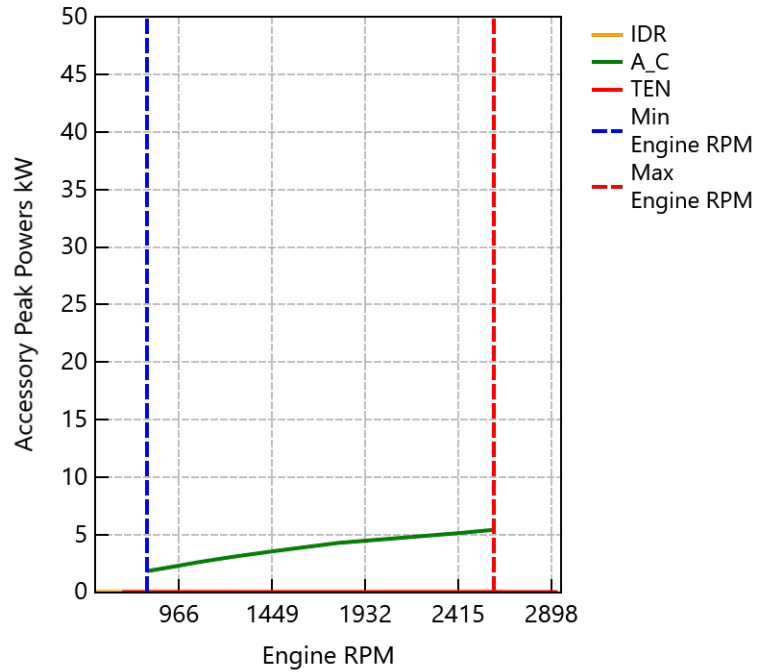
Tensioner T38624; CW; 24.6Nm

8PK1475HD Fleetranner

07Sep2023

Design Tension N	766 (172 [lbf])
Engine Acceleration RPM/s	1000
Engine Deceleration RPM/s	1000

Accessory Peak Powers kW



RESULTS

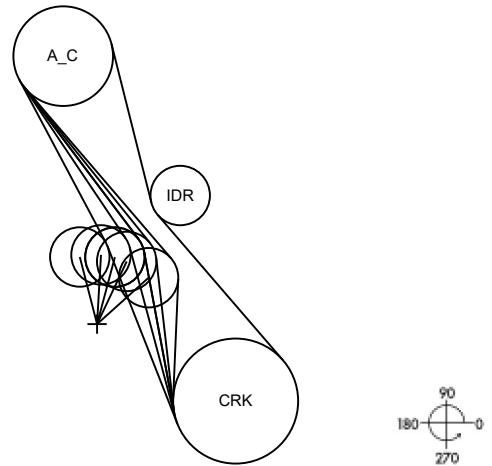
Peak Tension & Hubload

	Engine RPM	Accel. RPM/s	Tension N	Hubload N	Direction /Wrap °	Shear Stress N/mm-rib @ RPM
CRK	1262	1000	1113	1816.5	119 / 210	0.150 @ 1262
IDR	1262	1000	1110	512.6	28 / 27	N/A
A_C	1262	1000	810	1883.0	294 / 203	0.170 @ 1262
TEN	800	1000	766	362.0	211 / 26	N/A

Note: Peak Loads are defined as the highest loads (tension, shear stress and hubload) calculated for all peak load & accelerations conditions. (Not from system vibration calculation)



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Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	CRK	IDR	A_C	TEN
27.00	800	1	92	3.02	0.01	3.00	0.01
10.00	1000	1	92	4.02	0.01	4.00	0.01
13.00	1250	1	92	5.02	0.01	5.00	0.01
18.00	1500	1	92	5.52	0.01	5.50	0.01
19.00	1750	1	92	6.82	0.01	6.80	0.01
13.00	2000	1	92	8.22	0.01	8.20	0.01

Mean Hubloads N for DC 95%

%	Eng RPM	CRK	IDR	A_C	TEN
27.00	800	1911	557	1938	350
10.00	1000	1939	571	1967	350
13.00	1250	1939	571	1966	349
18.00	1500	1900	552	1927	349
19.00	1750	1925	564	1952	349
13.00	2000	1949	576	1977	349

Mean Tensions N for DC 95%

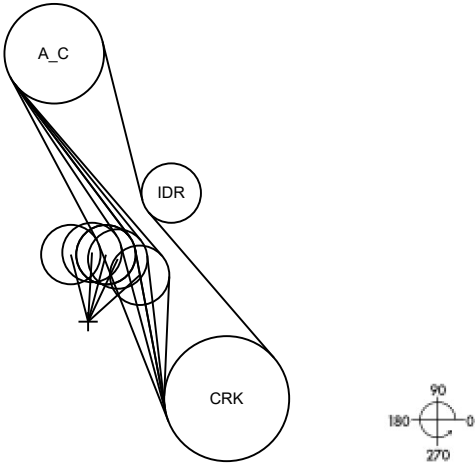
%	Eng RPM	CRK	IDR	A_C	TEN
27.00	800	1210	1208	767	766
10.00	1000	1238	1237	767	766
13.00	1250	1238	1237	767	766
18.00	1500	1198	1198	766	766
19.00	1750	1224	1223	766	766
13.00	2000	1249	1248	766	766



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Load Conditions [kW]	
Belt Data, DB Ver. = 2.18.0.0, Belt Rev. = 7/14/2016	
Belt Code	MT610
# of Ribs / Cord Material	8 / polyester

Tensioner Data	
Type	Automatic
Design Tension N	766 (172 lbf)





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RESULTS

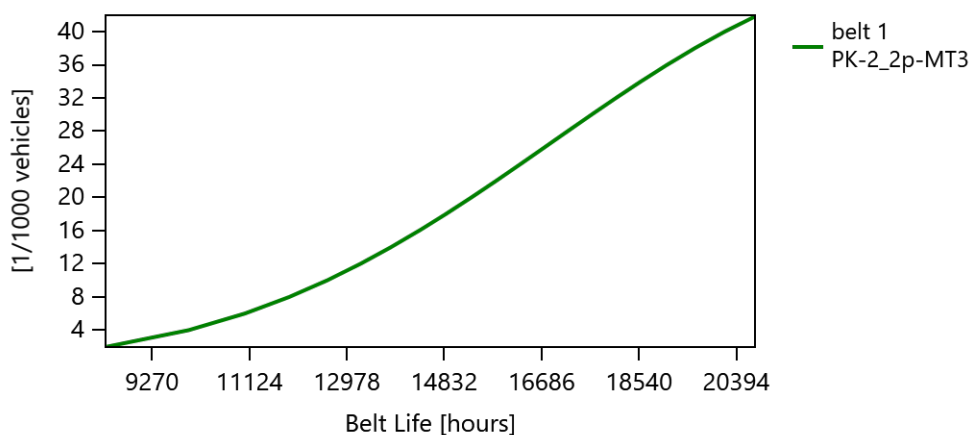
Pulley Contributions to Belt Rib Fatigue %

	CRK	IDR	A_C	TEN
PK-2_2p-MT3	2.9	43.3	10.5	43.3

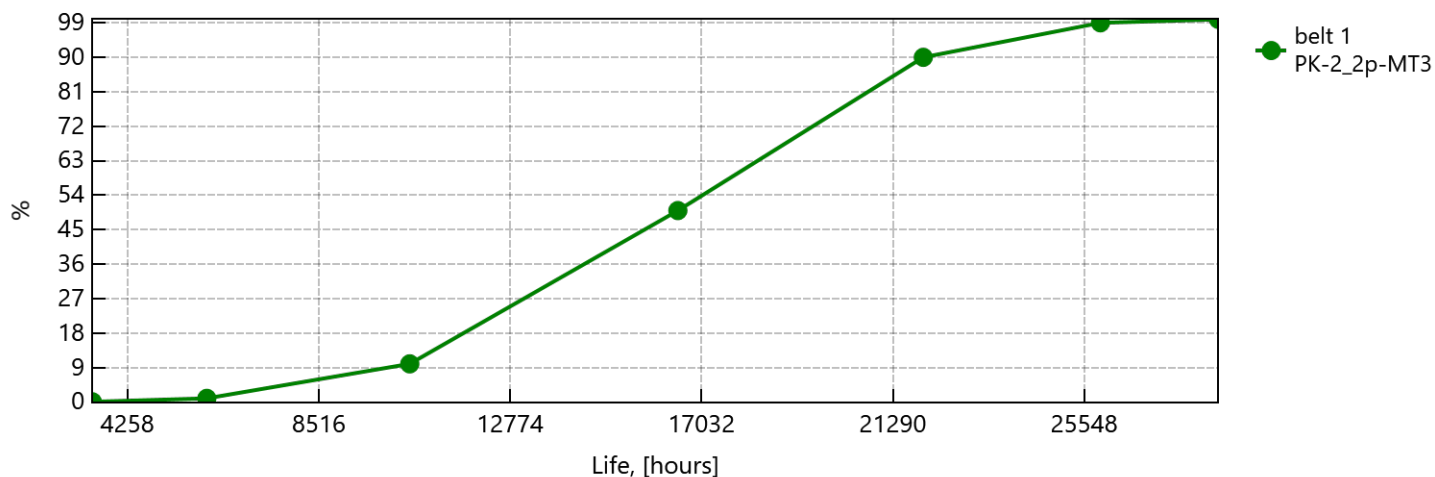
Belt Life B10

Duty Cycle DCn-th %	Av. Vehicle Speed km/hr	PK-2_2p-MT3 hours
95	1	10540

Belt Incidents per Thousand Vehicles



Weibull Distribution for DC95%



Load Conditions kW for DC 95%

%	Eng. RPM	V, km/hr	T, °C	CRK	IDR	A_C	TEN
27.00	800	1	92	3.02	0.01	3.00	0.01
10.00	1000	1	92	4.02	0.01	4.00	0.01
13.00	1250	1	92	5.02	0.01	5.00	0.01
18.00	1500	1	92	5.52	0.01	5.50	0.01
19.00	1750	1	92	6.82	0.01	6.80	0.01
13.00	2000	1	92	8.22	0.01	8.20	0.01

Load Contribution to Belt Rib Fatigue for DC 95%

%	Vblt m/s	CRK Tt N	CRK Ts N	% PK-2_2p-MT3
27.00	6.8	1210	766	17
10.00	8.5	1238	766	8
13.00	10.6	1238	766	13
18.00	12.8	1198	766	19
19.00	14.9	1224	766	24
13.00	17.0	1249	766	19



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Belt Code	MT610
# of Ribs / Cord Material	8 / polyester

Tensioner Data

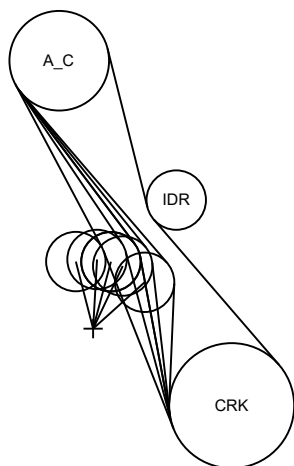
Type	Automatic
Design Tension N	766 (172 lbf)
Inertia of the Tensioner Arm Kg-m ²	0.0076
Tensioner Pulley Mass kg	0.50

Engine Data

# of Cylinders	6
Crankshaft Moment of Inertia kg m ²	0.1500

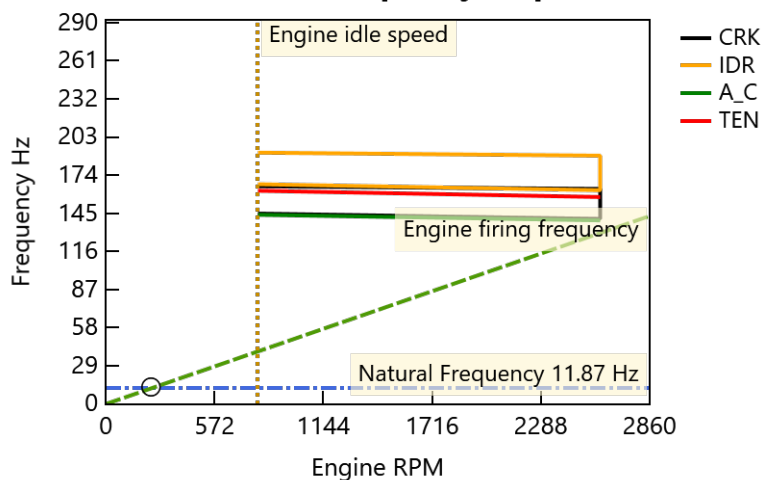
Accessory Data

	File Name - if used	Moment of Inertia kg m ²
IDR	IDR.cmp	0.0004
A_C	A_C.cmp	0.0050
TEN	TEN.cmp	0.0076



RESULTS

Natural Frequency Map



Note: Only low frequency spans are shown.
System Resonance (Mode 1: 12.61)

Note: If Engine Idle is near system resonant RPM,
then accessory drive system may have
excessive vibration.
If system resonance is within a span
frequency window, then span vibration
may have excessive amplitude.



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Drive Notes

Date	Note
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